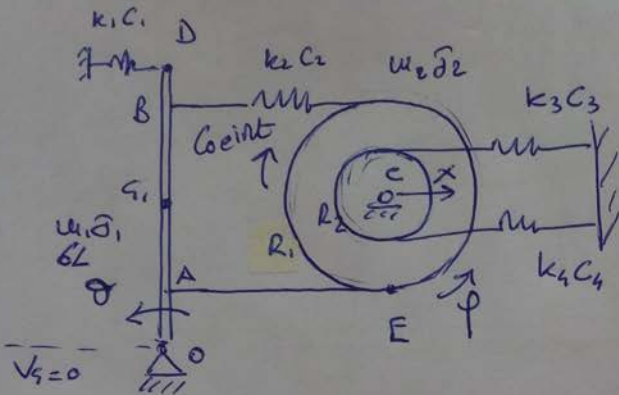


Written Test July 10th 2020

- Simplified problem -



2 rigid bodies $\rightarrow$ 6 dof
1 hinge, 1 cart and 1 string $\rightarrow$
 $\rightarrow$ 4 d.o.f.

→ 2 dof $x = \begin{Bmatrix} \delta \\ x \end{Bmatrix}$

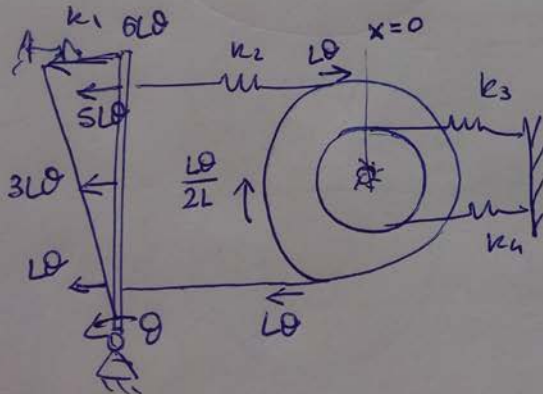
$$R_1 = 2L ; R_2 = L$$

$$T = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} I_1 \dot{\theta}_1^2 + \frac{1}{2} m_2 \dot{x}^2 + \frac{1}{2} I_2 \dot{\psi}^2$$

$$D = \frac{1}{2} \sum_{i=1}^4 c_i \dot{\Delta}_i^2 ; \quad V_{el} = \frac{1}{2} \sum_{i=1}^4 k_i \Delta_i^2 ; \quad V_g = m_i g h_{g_i}$$

($h_{g2} = \text{const}$)

$$S_W = -\text{Coef} \cdot S_P$$

$$\partial \neq 0; x=0:$$


$$\varphi = -\theta/2$$

$$\Delta L_1 = -660$$

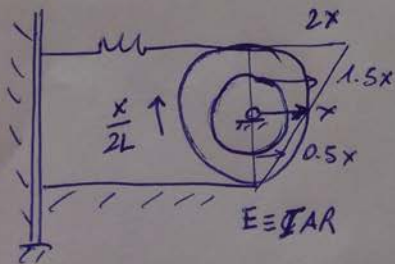
$$\Delta L_2 = 5L\theta + L\theta = 6L\theta$$

$$\Delta u_3 = \phi L = -\frac{\theta}{2} L$$

$$\Delta L_2 = -\phi L = \frac{\theta}{2} L$$

$$X_{G1} = -360$$

$g \neq 0 \times f \neq 0$



$$\phi = -\frac{x}{2L}$$

ΔL_{20}

$$\Delta c_2 = 2x$$

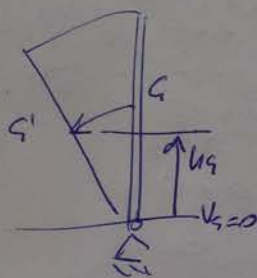
$$\Delta l_3 = -\frac{3}{2} \times$$

$$A_4 = -\frac{1}{2}x$$

$$x_{51} = 0$$

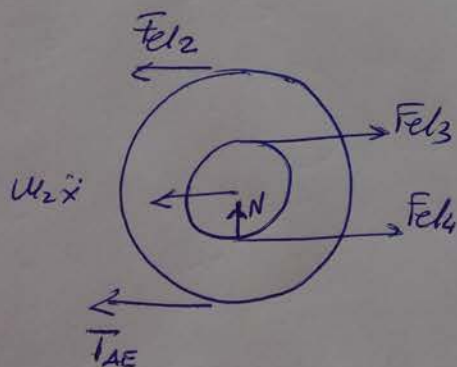
$$\begin{cases} \varphi = -\frac{\theta}{2} - \frac{x}{2L} \\ \Delta L_1 = -6L\theta \\ \Delta L_2 = 6L\theta + 2x \\ \Delta L_3 = -\frac{\theta}{2}L - \frac{3}{2}x \\ \Delta L_4 = \frac{\theta}{2}L - \frac{1}{2}x \\ X_{GA} = -3L\theta \end{cases} \quad [A_m] = \begin{bmatrix} -3L & 0 \\ 1 & 0 \\ 0 & 1 \\ -\frac{1}{2} & -\frac{1}{2L} \end{bmatrix}$$

$$[A_k] = [A_c] = \begin{bmatrix} -6L & 0 \\ 6L & 2 \\ -4/2 & -3/2 \\ 4/2 & -1/2 \end{bmatrix} \quad [A_Q] = \begin{bmatrix} -\frac{1}{2} & -\frac{1}{2L} \end{bmatrix}$$



$$u_g = 3L \cos \theta \rightarrow \frac{\partial^2 u_g}{\partial \theta^2} = -3L \cos \theta$$

$$[K_g] = \omega_{AS} \begin{bmatrix} -3L & 0 \\ 0 & 0 \end{bmatrix}$$



$$\sum F_x = 0 : T_{AE} = F_{el3} + F_{el4} - F_{el2} - \omega_2 \dot{x}'$$